RESEARCHLY AI: AI-POWERED RESEARCH ASSISTANT AND COLLABORATION PLATFORM FOR UNIVERSITY STUDENTS

COLLABORATION AND RECOMMENDATION ENGINE: SEMANTIC SUPERVISOR MATCHING, PEER CONNECT AND FEEDBACK ANALYTICS

Individual Thesis

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Supervisor allocation in universities is still driven largely by availability and keyword lists, collaborator discovery depends on networks undergraduates do not yet have, and supervision feedback is collected without aspect-level analysis. This thesis presents the Collaboration and Recommendation Engine (Module 2) of Researchly AI, an integrated research platform built for SLIIT. The module matches a student's free-text proposal against a directory of 74 SLIIT supervisors with a Sentence-BERT bi-encoder fine-tuned by contrastive learning on 147 labelled pairs, ranks candidates with a multi-factor score combining similarity (70%) and supervision capacity (30%) and explains each match; enriches profiles with live Semantic Scholar publications and a Gemini-generated research-focus summary; supports research-group formation; collects OTP-verified ratings with four-aspect sentiment analysis; and computes a transparent, re-normalised supervisor-effectiveness score. The module follows Design Science Research and is deployed as a FastAPI micro-service behind an authenticated gateway with a Next.js interface.

The matcher reached validation accuracy 0.973 and F1 0.986 during fine-tuning and, at a 0.60 threshold on 147 pairs, accuracy 0.946, precision 0.941, recall 1.000 and F1 0.970, exceeding the targets of 85% accuracy and 0.75 F1; live tests returned topically correct top-five supervisors for AI, cyber-security and NLP proposals. Of 13 functional validation cases, 10 passed, 2 were partial and 1 failed: zero-shot aspect sentiment collapsed to neutral on mixed feedback. Limitations include evaluation overlap with training data, a compressed similarity scale and a structural cap on directory supervisors' effectiveness scores.

Keywords: supervisor recommendation, Sentence-BERT, contrastive fine-tuning, aspect-based sentiment analysis, academic collaboration

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LIST OF ABBREVIATIONS

Abbreviation	Description
ABSA	Aspect-Based Sentiment Analysis
AI	Artificial Intelligence
AP	Average Precision
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CPU	Central Processing Unit
CRUD	Create, Read, Update, Delete
DSR	Design Science Research
F1	Harmonic mean of precision and recall
FR	Functional Requirement
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
MCC	Matthews Correlation Coefficient
ML	Machine Learning
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OTP	One-Time Password
PDPA	Personal Data Protection Act (Sri Lanka, No. 9 of 2022)
RDA	Research and Digital Archive (SLIIT institutional repository)
REST	Representational State Transfer
RLS	Row-Level Security
SBERT	Sentence-BERT
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SMTP	Simple Mail Transfer Protocol
SO	Specific Objective
STS	Semantic Textual Similarity
SUS	System Usability Scale
UAT	User Acceptance Testing

Abbreviation	Description
UI	User Interface
URL	Uniform Resource Locator

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1 INTRODUCTION

1.1 Background and Overall Project Context

University students face a research workflow that has grown far faster than the tools supporting it. Global scholarly output rose from roughly 2.1 million articles in 2010 to more than 3.3 million in 2022 [1], and a final-year undergraduate must survey that literature, select a defensible topic, find a suitable supervisor, cite sources correctly, avoid overlap with earlier work and write a structured proposal, usually within a single academic year and with limited supervision time. Each of these tasks is served by a separate tool — a reference manager, a search engine, a similarity checker, a spreadsheet of supervisors — and none of these tools share data [2], [3].

The group project R26-IT-116, **Researchly AI — an AI-powered research assistant and collaboration platform for university students**, addresses this fragmentation by placing four intelligent modules behind one web application and one shared data substrate: Module 1 (Research Integrity and Compliance), Module 2 (Collaboration and Recommendation), Module 3 (Research Data Collection and Management) and Module 4 (Research Performance Analytics and Visualization). All four modules are grounded in a corpus of 4,219 records drawn from SLIIT's institutional research repository (RDA), of which 3,858 carry usable abstracts, so that every recommendation, gap, forecast or similarity score refers back to real prior SLIIT research rather than to generic web content. The platform is deployed as a Next.js front end on Vercel, an Express.js API gateway and four FastAPI micro-services, backed by Supabase PostgreSQL with the pgvector extension (Figure 1.1).

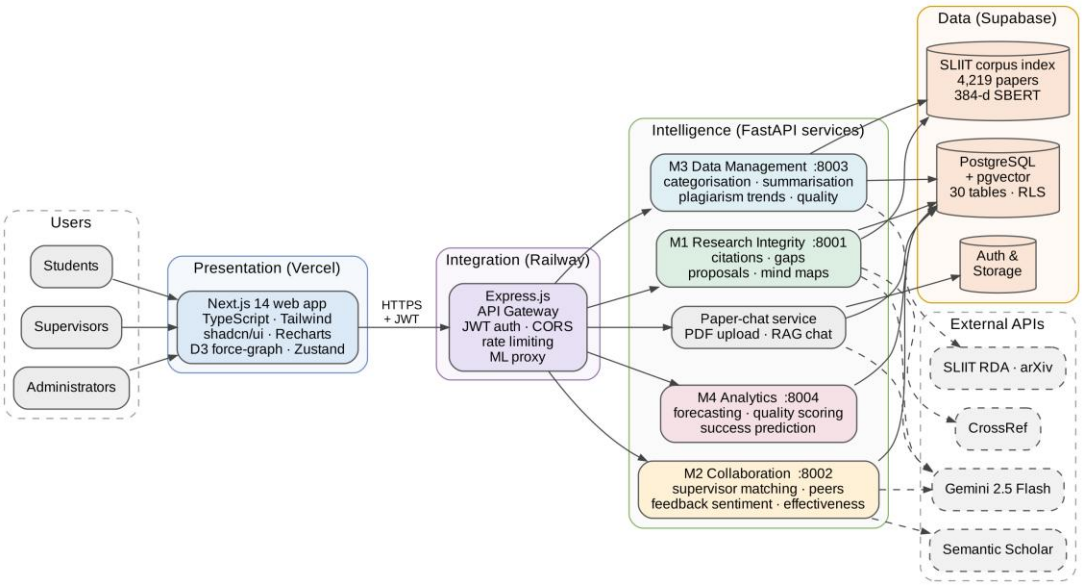


Figure 1.1: High-level architecture of the Researchly AI platform showing the presentation, integration, intelligence and data layers

This thesis reports the author's individual component, **Module 2: Collaboration and Recommendation Engine**. The common group report describes the integrated solution, the shared infrastructure and the end-to-end evaluation; this document concentrates on the research problem, design, implementation and validation of the individual component, and on how it exchanges data with the other three modules.

1.2 Individual Component Context

Module 2, the **Collaboration and Recommendation Engine**, is the people-facing part of the platform. Research outcomes depend not only on what a student studies but on *who* they work with: a supervisor whose expertise fits the topic, peers with complementary skills, and a feedback loop that tells the institution which supervision practices work. The module is implemented as the FastAPI service `services/module2-collaboration` (port 8002) and four pages under `/collaboration` in the web application:

- **Supervisor Matching** — ranks the 74 supervisors of the SLIIT directory against a student's free-text proposal using a Sentence-BERT bi-encoder and a multi-factor score that accounts for availability, and explains each match.

- **Supervisor Profile Enrichment** — shows a supervisor's real publication history from the Semantic Scholar API [4], a year-distribution chart, a research-topic chart and an AI-generated summary of their recent research focus.
- **Peer Connect** — lets students publish research groups with open slots and request to join, with a content-based SBERT peer matcher retained for interest-based discovery.
- **Feedback and Aspect-Based Sentiment** — collects e-mail-verified star ratings and written feedback and analyses sentiment for four aspects (methodology, writing, originality and data analysis).
- **Supervisor Effectiveness** — aggregates ratings, sentiment, satisfaction and completion rate into a transparent, re-normalised effectiveness score.

The module shares the embedding model family and the gateway with the other modules; confirmed supervisor matches are recorded in the shared `supervisor_matches` table.

1.3 Problem Statement

In most departments, supervisor allocation is driven by availability, acquaintance and keyword lists rather than by demonstrable alignment of research interests. Supervisor–student fit and continuity of supervision are associated with student progress and outcomes [5], yet a student choosing among 74 supervisors across eleven departments has no systematic way to compare a proposal against each supervisor's expertise. Keyword matching fails precisely when a student and a supervisor describe the same problem in different words (for example, "crop monitoring with edge devices" versus "wireless sensor networks"). Academic recommender systems overwhelmingly recommend *papers* rather than *people* [3], and collaborator recommenders rely on co-authorship networks that undergraduates do not yet have [6], [7]. Finally, feedback on supervision is collected — if at all — as star ratings whose free-text comments are never analysed by aspect [8].

The problem addressed by this component is therefore: **students lack an explainable, semantics-aware way to identify suitable supervisors and collaborators, and institutions lack a trustworthy, aspect-level view of supervision quality.**

1.4 Research Gap

Table 1.1: Research gaps addressed by Module 2

ID	Gap	Evidence
G1	Academic recommenders focus on papers; supervisor allocation is rarely modelled.	Surveys [3], [9]; supervisor selection [10]
G2	Collaborator recommenders depend on co-authorship graphs unavailable for undergraduates.	[6], [7], [11]
G3	Keyword matching misses semantically equivalent research descriptions.	SBERT [12]; STS [13]
G4	Student feedback is analysed at document level, not by aspect of supervision.	ABSA [14], [15]; education survey [8]
G5	Teaching-evaluation scores are opaque and conflate signals.	[16], [17]

1.5 Aim and Objectives

Aim. To design, implement and evaluate an AI-driven Collaboration and Recommendation Engine that matches students with suitable supervisors and peers through embedding-based semantic similarity, analyses academic feedback by aspect and computes transparent supervisor-effectiveness scores, as an integrated component of the Researchly AI platform.

- **SO1** — Implement a Sentence-BERT matching algorithm between student proposals and supervisor expertise profiles with a matching accuracy of at least 85% (F1 at least 0.75 on labelled pairs).
- **SO2** — Develop a peer collaboration feature that helps students find collaborators with complementary interests.
- **SO3** — Build a feedback sentiment analyser that classifies academic feedback by aspect.
- **SO4** — Create a supervisor-effectiveness score combining ratings, sentiment, satisfaction and completion.

- **SO5** — Develop a multi-factor ranking that combines similarity with availability and explains each recommendation.
- **SO6** — Integrate the module into the platform through the API gateway and shared data layer.

1.6 Research Questions

- **RQ1:** How accurately does a fine-tuned Sentence-BERT bi-encoder match free-text student proposals to supervisor expertise profiles, and is the ranking topically correct across research domains?
- **RQ2:** Can a zero-shot large language model provide usable aspect-level sentiment for supervision feedback without a labelled training set, and how does it fail?
- **RQ3:** Does a transparent weighted aggregation provide a defensible supervisor-effectiveness indicator, and what structural biases does it introduce?

1.7 Scope, Limitations and Assumptions

In scope: the 74-supervisor SLIIT directory and system-registered supervisors; English proposals; group formation by self-publication; four supervision aspects; e-mail-verified ratings. **Out of scope:** automatic allocation decisions (the system recommends, staff decide), co-authorship network analysis, and multilingual feedback. **Assumptions:** directory research interests and keywords are accurate and current; students describe their topic in at least one or two sentences; verified e-mail addresses identify genuine raters.

1.8 Individual Contribution

The author designed and implemented the Module 2 service — routers `supervisor.py` (395 lines), `peer.py` (364), `feedback.py` (452) and `effectiveness.py` (279), and the services `supervisor_matcher.py` and `supervisor_directory.py` — compiled the SLIIT supervisor dataset (74 profiles with departments, clusters, interests, keywords and capacity) and the 147 labelled training pairs, wrote the fine-tuning and evaluation scripts

(`train_supervisor_matcher.py`, `generate_more_pairs.py`, `evaluate_model.py`, `upload_to_supabase.py`), designed migration `013_peer_groups_and_supervisor_ratings.sql`, and built the four collaboration pages of the front end.

1.9 Thesis Structure

Chapter 2 reviews related work; Chapter 3 describes the methodology; Chapter 4 presents requirements and design; Chapter 5 covers implementation, integration and testing; Chapter 6 reports results; Chapter 7 discusses them; Chapter 8 addresses commercialisation; and Chapter 9 concludes, following the SLIIT guidelines [18].

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

IEEE Xplore, ACM Digital Library, Scopus-indexed journals (via Google Scholar) and Semantic Scholar were searched for 2010–2026 work using *supervisor recommendation*, *collaborator recommendation*, *academic recommender systems*, *semantic textual similarity*, *contrastive sentence embeddings*, *aspect-based sentiment analysis*, *student feedback sentiment* and *teaching evaluation validity*. Studies were included if they proposed or evaluated a technique for matching people in academia, embedding-based similarity, or feedback analysis.

2.2 Domain and Theoretical Foundations

2.2.1 Recommender systems in academia

Recommender systems are typically content-based, collaborative or hybrid [9]. Beel et al. reviewed more than 200 research-paper recommender articles and found content-based filtering dominant and evaluation practices inconsistent [3]. Haruna et al. proposed collaborative filtering for paper recommendation using the citation graph [11]. For *people*, Kong et al. combined dynamic research interests with academic influence to recommend collaborators [6], and Xi et al. combined interest similarity with network topology [7]. Both assume publication and co-authorship histories, which undergraduates lack. Zhang et al. proposed a research-analytics framework for supervisor selection based on supervisors' publications [10], the closest prior work to this module.

2.2.2 Semantic similarity with sentence embeddings

Semantic textual similarity measures how close two texts are in meaning [13]. BERT cross-encoders achieve high accuracy but require a forward pass per pair [19]; Sentence-BERT bi-encoders embed each text once so that similarity reduces to cosine similarity between vectors [12]. For matching, this means supervisor embeddings can be computed offline and each student query requires one forward pass. MiniLM distillation [20] yields compact 384-dimensional models such as `all-MiniLM-L6-v2` that run on CPU. Contrastive objectives pull matching pairs together and push non-

matching pairs apart in embedding space [21], which is the basis for domain fine-tuning.

2.2.3 Aspect-based sentiment analysis

Aspect-based sentiment analysis (ABSA) assigns polarity to specific aspects mentioned in text [14]. BERT-based ABSA reformulates the task as sentence-pair classification with auxiliary sentences [15]. Shaik et al. survey sentiment analysis on educational data and highlight the scarcity of labelled datasets and the domain shift between product reviews and academic feedback [8]. Zhang et al. show that large language models perform well zero-shot on simpler sentiment tasks but lag on structured, fine-grained tasks such as ABSA [22].

2.2.4 Measuring supervision and teaching effectiveness

Student evaluations of teaching can be reliable when instruments are multidimensional [16], but Bedggood and Donovan argue that aggregate scores often measure satisfaction rather than learning [17]. Ives and Rowley found that supervisor selection and continuity affect doctoral progress [5]. These works caution that any effectiveness score must expose its components and weights rather than present a single opaque number.

2.3 Related Research

Table 2.1: Related research informing Module 2

Study	Approach	Relevance and limitation
Zhang et al. [10]	Analytics-supported supervisor selection	Uses publication analytics; no semantic matching of free-text proposals.
Kong et al. [6]	Dynamic interests + academic influence	Needs publication history; not applicable to undergraduates.
Xi et al. [7]	Interests + network topology	Co-authorship network required.
Beel et al. [3]	Survey of paper recommenders	Content-based dominant; people recommendation rare.
Reimers and Gurevych [12]	Siamese sentence embeddings	Efficient pre-computed similarity; core technique of this module.
Hadsell et al. [21]	Contrastive loss	Basis for fine-tuning on positive/negative pairs.
Pontiki et al. [14]	ABSA benchmark	Defines aspect polarity task used for feedback.

Study	Approach	Relevance and limitation
Zhang et al. [22]	LLMs for sentiment	Zero-shot LLMs viable but weaker on fine-grained tasks.
Shaik et al. [8]	Survey of educational sentiment	Scarce labelled data; domain shift.

2.4 Existing Systems, Products and Techniques

Table 2.2: Comparison of existing products with Module 2

Product	Supervisor matching	Peer discovery	Feedback analysis	Effectiveness view
ResearchGate / Google Scholar profiles	Manual browsing	Follow / message	No	Citations only
University staff directory	Keyword list	No	No	No
Course evaluation forms	No	No	Overall score	Aggregate only
LinkedIn-style networks	No	Social graph	No	No
Module 2	Semantic + availability, explained	Group marketplace + SBERT	Aspect-level sentiment	Transparent weighted score

2.5 Gap Synthesis and Conceptual Framework

The review points to a design that (i) uses a bi-encoder so supervisor vectors are pre-computed and each query is cheap, (ii) adapts the encoder to SLIIT phrasing with contrastive fine-tuning, (iii) blends similarity with real-world capacity so recommendations are actionable, (iv) explains every match, and (v) treats feedback analysis and effectiveness scoring as transparent aggregations of verifiable signals. Figure 2.1 places these features in the student journey.

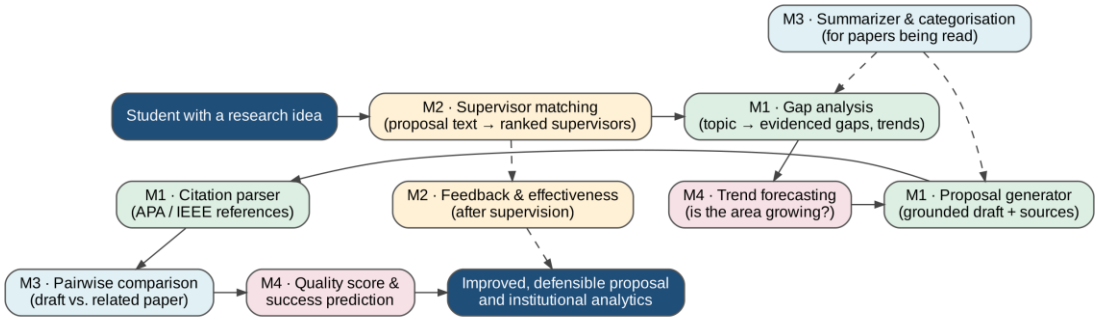


Figure 2.1: Student journey through the integrated platform; Module 2 supports supervisor matching at the start and feedback after supervision

3 RESEARCH METHODOLOGY

3.1 Research Design

The work follows Design Science Research [23], [24]: the matcher, group marketplace, feedback analyser and effectiveness score were designed as artefacts, built iteratively in two-week sprints and evaluated against the objectives using quantitative metrics on labelled pairs and functional validation through the live service.

3.2 Data, Participants and Study Materials

Table 3.1: Data and study materials used by Module 2

Material	Description	Use
SLIIT supervisor directory	74 supervisors: rank (9 Prof., 17 Dr., 28 Ms., 20 Mr.), department, research cluster, 3.9 research interests on average, keywords, availability and capacity (225 slots, 91 in use)	Matching corpus; directory for ratings
Training pairs	147 labelled (proposal, supervisor, label) pairs: 127 positive, 20 negative, augmented with synonym substitution and hard negatives	Contrastive fine-tuning and evaluation
Semantic Scholar Graph API	Live author search and publication lists	Profile enrichment
Feedback test texts	Clearly positive, clearly negative and mixed feedback passages	Validation of sentiment analysis
Supabase tables	peer_groups, peer_group_join_requests, supervisor_ratings, supervisor_matches	Groups, ratings and effectiveness

No personal data of real students was used for training. Supervisor information was compiled from the public SLIIT staff directory. Validation used synthetic feedback passages.

3.3 Experimental Procedure

1. Compile and clean the supervisor directory and write its JSON and Supabase representations (`sliit_supervisors.json`, `supabase_migration.sql`).
2. Construct labelled proposal–supervisor pairs and augment them (`generate_more_pairs.py`).

3. Fine-tune `all-MiniLM-L6-v2` with contrastive loss for 15 epochs (batch 16, learning rate 2×10^{-5}, 10 warm-up steps, 15% validation split, seed 42) and pre-compute the 74 supervisor embeddings (`train_supervisor_matcher.py`).
4. Evaluate binary match classification at threshold 0.60 and inspect qualitative top-5 lists for seven domain queries (`evaluate_model.py`).
5. Execute the 13-case functional validation through the running service on 28 August 2026 and record verdicts.

3.4 Measures and Evaluation Metrics

Table 3.2: Evaluation measures for Module 2

Objective	Measure	Target
SO1	Accuracy, precision, recall, F1, average precision on labelled pairs; topical correctness of top-5 for domain queries	Accuracy ≥ 85%, F1 ≥ 0.75
SO2	Correct creation, listing and join-request flow	Functional
SO3	Correct polarity for positive, negative and mixed feedback per aspect	Differentiated aspects
SO4	Hand-verification of formula on real ratings	Exact agreement
SO5	Ranking stability and explanation quality	Qualitative
SO6	Routes reachable through gateway with auth	All routes

3.5 Data Analysis

Classification metrics were computed from the confusion matrix of predicted versus labelled pairs at a cosine threshold of 0.60. Training curves were read from the Sentence-Transformers evaluator log. Ranking relevance was judged by comparing each recommended supervisor's declared interests with the query domain. Validation verdicts follow the PASS/PARTIAL/FAIL/BLOCKED scheme with severity and root cause.

3.6 Ethics, Privacy, Security and Responsible AI

- The corpus consists only of publicly available bibliographic records and abstracts from SLIIT's open research repository; full texts uploaded by users

are stored in a private Supabase Storage bucket protected by Row-Level Security [25].

- Personal data (names, e-mail addresses, research interests, ratings) are processed in line with the principles of Sri Lanka's Personal Data Protection Act No. 9 of 2022 — purpose limitation, data minimisation and access control [26].
- Every request to an intelligence service passes through the gateway, which verifies a Supabase-issued JSON Web Token [27] and applies rate limiting (600 requests per minute globally and 200 per minute for ML routes).
- Generative components (Gemini) are used only where interpretation is genuinely required and every retrieval-based output is traceable to a cited SLIIT paper, which reduces the risk of hallucinated content [28].
- Ratings require a verified e-mail address (one-time code) to prevent fake reviews, and rater identity is stored only with the rating for accountability.
- Effectiveness scores expose their component values and weights so that supervisors and administrators can see why a score is high or low; the score is presented as decision support, not as a performance appraisal.
- Supervisor recommendations are advisory: allocation decisions remain with academic staff.

3.7 Reproducibility and Research Artefacts

Training data (`data/training_pairs.json`), the supervisor dataset, the training and evaluation scripts, the evaluator log (`eval/binary_classification_evaluation_supervisor_val_results.csv`) and the evaluation output (`data/evaluation_results.json`) are versioned in the repository. The validation harness `docs/module-reports/_test_module2.py` reproduces the functional tests.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1: Functional requirements of Module 2

ID	Requirement
FR1	Accept a free-text proposal (or legacy interests + abstract) and return the top-K supervisors with similarity, multi-factor score, explanation and capacity.
FR2	Show a supervisor profile with publications, year distribution, topic distribution and an AI summary of recent focus.
FR3	Allow students to create research groups with open slots, browse open groups and send join requests to the leader.
FR4	Recommend peers with shared and complementary interests (legacy content-based matcher).
FR5	Verify a rater's e-mail with a one-time code before accepting a rating.
FR6	Accept 1–5 star ratings and written feedback and analyse sentiment for methodology, writing, originality and data analysis.
FR7	Compute and display supervisor-effectiveness scores with their breakdown and recent verified feedback.

Table 4.2: Non-functional requirements of Module 2

ID	Category	Requirement
NFR1	Performance	Supervisor matching within 10 s; sentiment analysis within 5 s (proposal targets).
NFR2	Scalability	Support at least 500 student and 100 supervisor profiles.
NFR3	Reliability	External API failures (Semantic Scholar, Gemini, SMTP) must not break the page.
NFR4	Security	JWT-authenticated routes; OTP-verified ratings; rate limits.
NFR5	Transparency	Similarity, weights and score components visible to users.
NFR6	Usability	Responsive pages; one-click profile and contact actions.

4.2 Component Architecture within the Complete Solution

Figure 4.1 shows the component. The gateway file `module2.routes.ts` defines 17 routes, all protected by `requireAuth` and `mlRateLimiter` and proxied with a 60-second timeout. Four routers delegate to the matcher, the supervisor directory, Supabase and two external services: the Semantic Scholar Graph API for publications and Gemini 2.5 Flash for sentiment and focus summaries. Two ML approaches coexist

deliberately: a local fine-tuned bi-encoder where scores must be deterministic and explainable, and a general LLM where open-ended interpretation is needed.

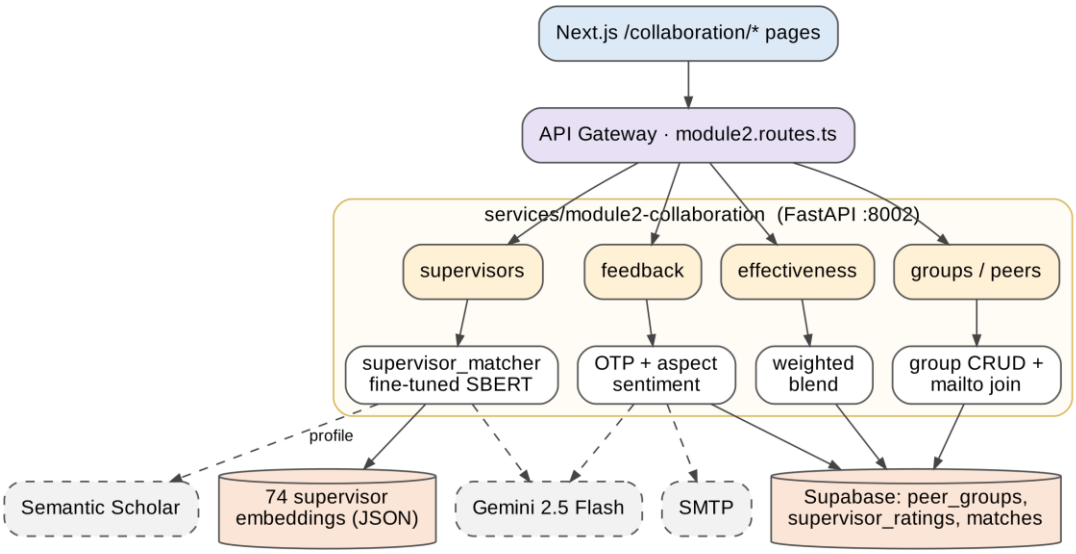


Figure 4.1: Component architecture of the Collaboration and Recommendation Engine

4.3 Design Alternatives and Decisions

Table 4.3: Design decisions and rationale for Module 2

Concern	Alternative considered	Decision	Rationale
Matching model	Cross-encoder or LLM ranking	Bi-encoder with pre-computed supervisor embeddings	One forward pass per query; CPU-friendly; scores are true cosine similarities [12].
Domain adaptation	Base model only	Contrastive fine-tuning on SLIIT pairs	Aligns SLIIT phrasing [21]; base model retained as fallback.
Ranking	Similarity only	0.70 × similarity + 0.30 × availability	Prevents full supervisors crowding out available ones.
Peer formation	Pure SBERT peer matching	Group marketplace + legacy matcher	Group formation depends on timing and chemistry as much as topic.
Feedback sentiment	Fine-tuned BERT ABSA (target ≥ 93%)	Gemini zero-shot JSON, star-rating fallback	No labelled SLIIT feedback dataset; avoids a separate model artefact [22].
Effectiveness	Learned model	Transparent weighted, re-normalised blend	Auditable; affects reputation [17].

4.4 Detailed Design

4.4.1 Supervisor matching

Figure 4.2 shows the matching pipeline. The router accepts either a `proposal` string or the legacy `research_interests[]` plus `abstract`, which are joined into one proposal. `load_model()` loads the fine-tuned checkpoint from `models/trained_supervisor_matcher/` when `model.safetensors` is present and otherwise falls back to the base `all-MiniLM-L6-v2` checkpoint. The query is encoded once; supervisor embeddings are read from `supervisors_with_embeddings.json`, so no supervisor text is re-encoded at query time. Cosine similarity is computed as $\text{dot}(a,b)/(\|a\|\|b\| + 10^{-8})$, candidates below the minimum similarity (default 0.2) are dropped and the remainder are scored:

$$\text{availability} = 0 \text{ if unavailable; } 0.3 \text{ if current_students} \geq \text{max_students; } 1.0 \text{ otherwise}$$

$$\text{multi_factor} = 0.70 \cdot \text{similarity} + 0.30 \cdot \text{availability}$$

An explanation is generated from a template chosen by whether any of the supervisor's interest keywords occurs in the query and by the similarity band (> 0.70 strong, > 0.55 could supervise, otherwise some relevant expertise). Results are sorted by the multi-factor score and truncated to top-K (default 5).

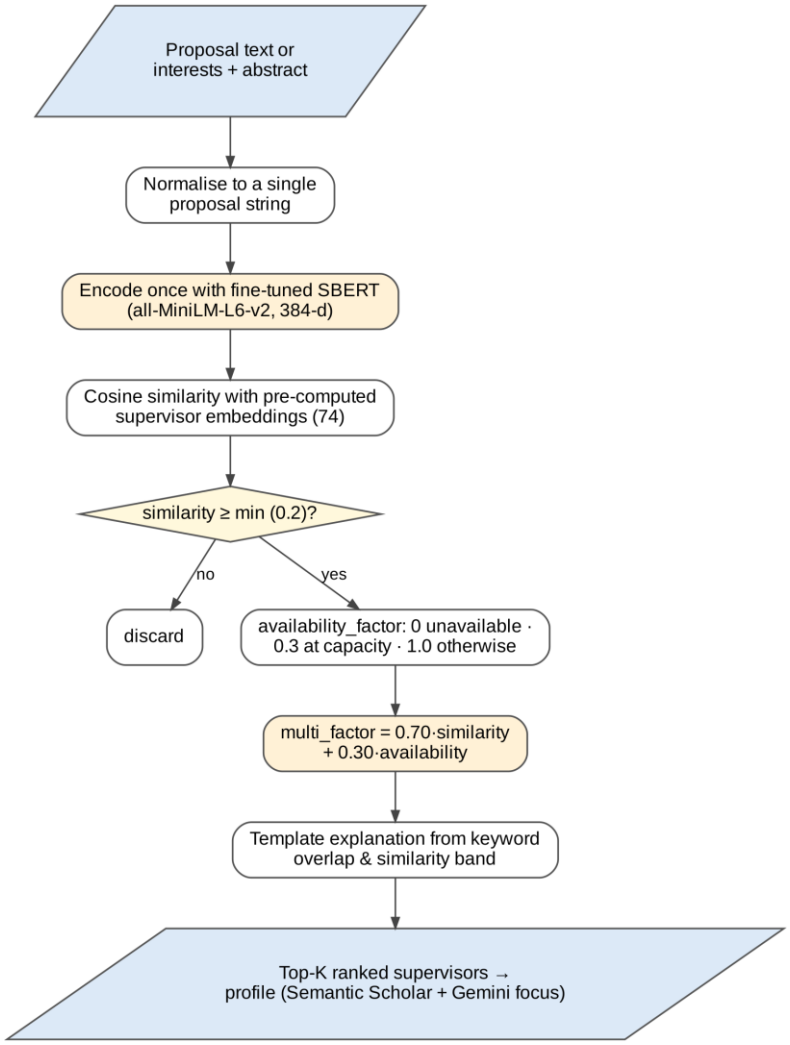


Figure 4.2: Supervisor matching pipeline with multi-factor ranking

4.4.2 Profile enrichment

A profile request searches Semantic Scholar by author name and then fetches the author's papers. Because the free tier allows one request per second per process, a module-level `asyncio.Lock` and a timestamp enforce a 1.05-second interval, and HTTP 429 responses are retried up to three times with increasing back-off. Successful lookups are cached for one hour and failures for 90 seconds, so that a transient rate limit does not hide a supervisor's data for an hour. Papers from 2010 onward populate a year bar chart and the supervisor's declared interests (top ten) a pie chart. The 15 most recent titles are sent to Gemini, which returns a JSON summary, recent focus

areas and an activity level; this summary is cached for 24 hours and omitted silently on failure.

4.4.3 Peer Connect

Students create a group (`POST /matching/groups`) with title, description, research area, members, slots and contact e-mail; open groups are listed (`GET /matching/groups`), and a join request (`POST /matching/groups/{id}/join-request`) is stored in `peer_group_join_requests` and returned with a pre-filled `mailto:` link so that the message is sent from the requester's own mailbox without a server-side relay. The legacy content-based matcher embeds a student's interests and calls the `match_peers` pgvector RPC with threshold 0.3, reporting shared interests (intersection) and complementary skills (set difference).

4.4.4 Feedback, OTP verification and aspect sentiment

Figure 4.3 shows the rating flow. A six-digit code with a ten-minute lifetime is stored in memory keyed by the lower-cased e-mail and sent by SMTP (SSL on port 465 or STARTTLS on 587); if SMTP is not configured, the code is returned in the response for demonstrations. Verification is single-use. When written feedback is provided, Gemini is asked for strict JSON with four aspect objects (sentiment and positive/neutral/negative probabilities) and an overall verdict; if the overall field is missing, it is derived from the mean aspect score (> 0.3 positive, < -0.3 negative). Without text, sentiment is derived from stars as $(\text{stars} - 3)/2$.

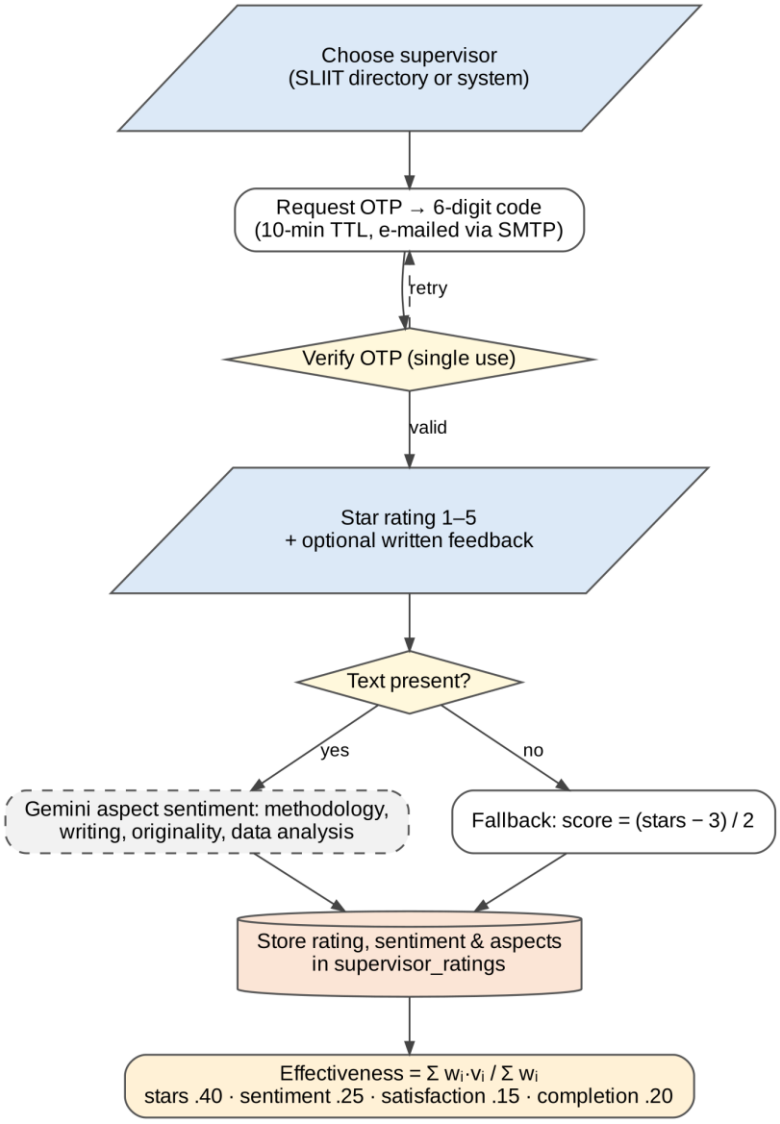


Figure 4.3: Verified rating, aspect sentiment and effectiveness pipeline

4.4.5 Supervisor effectiveness

Effectiveness combines four signals, each normalised to 0–1:

stars = avg_stars / 5; sentiment = (avg_sentiment + 1) / 2; satisfaction = share of ratings with sentiment > 0.3

completion = completed_matches / total_matches (system supervisors only)

overall = $\Sigma\, w_i \cdot v_i / \Sigma\, w_i$ over available signals, $w = (0.40, 0.25, 0.15, 0.20)$

Missing signals are removed from both numerator and denominator, so a supervisor with no ratings still receives a defined score from completion alone. The response exposes `breakdown.weights` and every component.

4.5 Standards and Professional Practice

- Quality attributes follow ISO/IEC 25010 categories [29]; web security follows OWASP Top 10 guidance [30].
- External API etiquette: rate limits and caching for Semantic Scholar [4].
- Personal data handling follows PDPA principles [26]; OTP codes are single-use and short-lived.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1: Technology stack

Layer	Technologies	Justification
Presentation	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, shadcn/ui, Recharts, D3 / react-force-graph-2d, Zustand, Axios	Server-side rendering, typed components, responsive layout and rich interactive charts in one framework [31]
Integration	Express.js API gateway (TypeScript), express-rate-limit, Supabase JWT verification, per-module proxy with 60 s / 150 s timeouts	Single entry point that authenticates once and isolates the front end from service locations [32]
Intelligence	Python 3.10+, FastAPI, sentence-transformers (all-MiniLM-L6-v2, fine-tuned), scikit-learn, XGBoost, statsmodels, PyMuPDF / pypdf / pdfplumber, Google Gemini 2.5 Flash client	Asynchronous typed micro-services; CPU-friendly models that fit the hosting budget [33], [34]
Data	Supabase PostgreSQL 15 + pgvector, Row-Level Security, Supabase Auth and Storage; pre-computed NumPy embedding matrices and pickled model bundles	Relational and vector search in one managed service, policy-based access control [35], [25]
DevOps	pnpm workspaces + Turborepo monorepo, Docker multi-stage images, GitHub Actions, Vercel, Railway, Azure Container Registry workflow, Playwright	Reproducible builds and automatic deployment on every push to main [36]

Module-specific dependencies include `sentence-transformers`, `torch` (CPU), `httpx` for Semantic Scholar, `smtplib` and `google-generativeai`.

5.2 Component Implementation

Table 5.2: Key Module 2 implementation files

File	Lines	Responsibility
app/routers/feedback.py	452	OTP request/verify, Gemini aspect sentiment, rating submission, directory
app/routers/supervisor.py	395	Matching endpoint, Semantic Scholar profile, Gemini focus summary, caching
app/routers/peer.py	364	Group CRUD, join requests with mailto, legacy peer matcher
training/train_supervisor_matcher.py	295	Contrastive fine-tuning, embedding export
app/routers/effectiveness.py	279	Weighted, re-normalised effectiveness score and detail view
app/services/supervisor_matcher.py	180	Model loading, cosine scoring, multi-factor ranking, explanations
training/evaluate_model.py	167	Threshold evaluation and qualitative top-5 lists

File	Lines	Responsibility
web: collaboration/supervisor-match/page.tsx	634	Matching UI and supervisor detail view with charts
web: collaboration/peer-connect/page.tsx	545	Browse/create groups, join dialog
web: collaboration/feedback/page.tsx	533	Three-step verified rating UI
web: collaboration/effectiveness/page.tsx	497	Effectiveness list and detail

The multi-factor score is implemented in a few lines, which keeps the policy easy to audit and change:

```
# Availability factor
avail_factor = 1.0
if not sup.get("availability", True):
    avail_factor = 0.0  # Unavailable
elif sup.get("current_students", 0) >= sup.get("max_students", 5):
    avail_factor = 0.3  # Full capacity

# Multi-factor score: 70% similarity + 30% availability
final_score = (sim * 0.70) + (avail_factor * 0.30)
```

Figures 5.1 to 5.5 show the implemented pages.

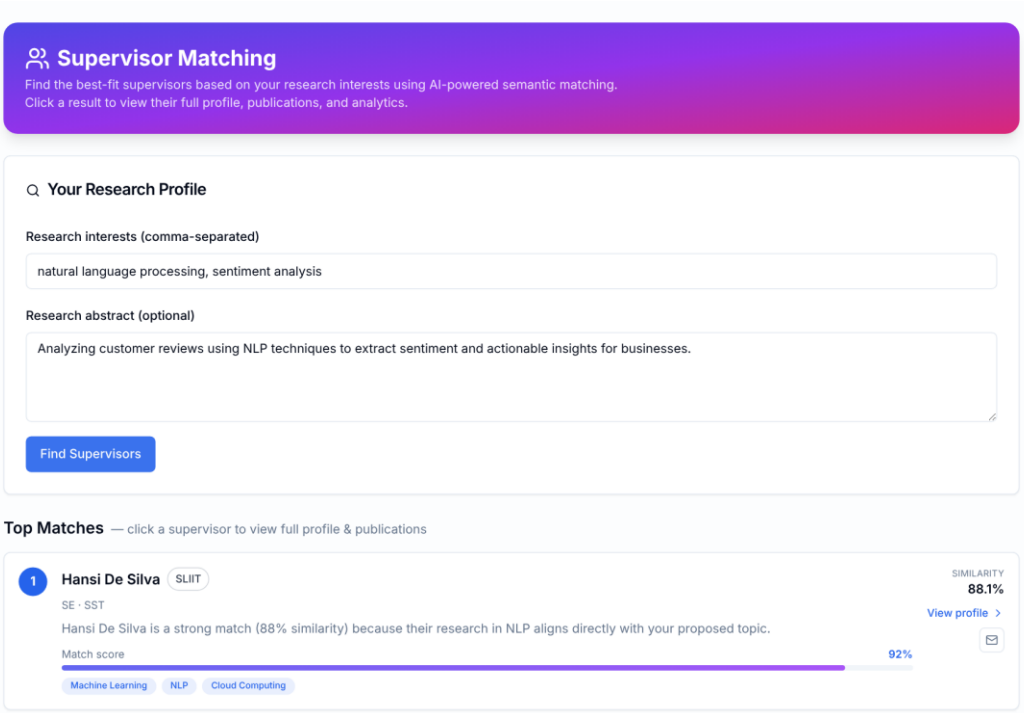


Figure 5.1: Supervisor Matching page: research interests and abstract entered, ranked matches with similarity, match score and expertise tags

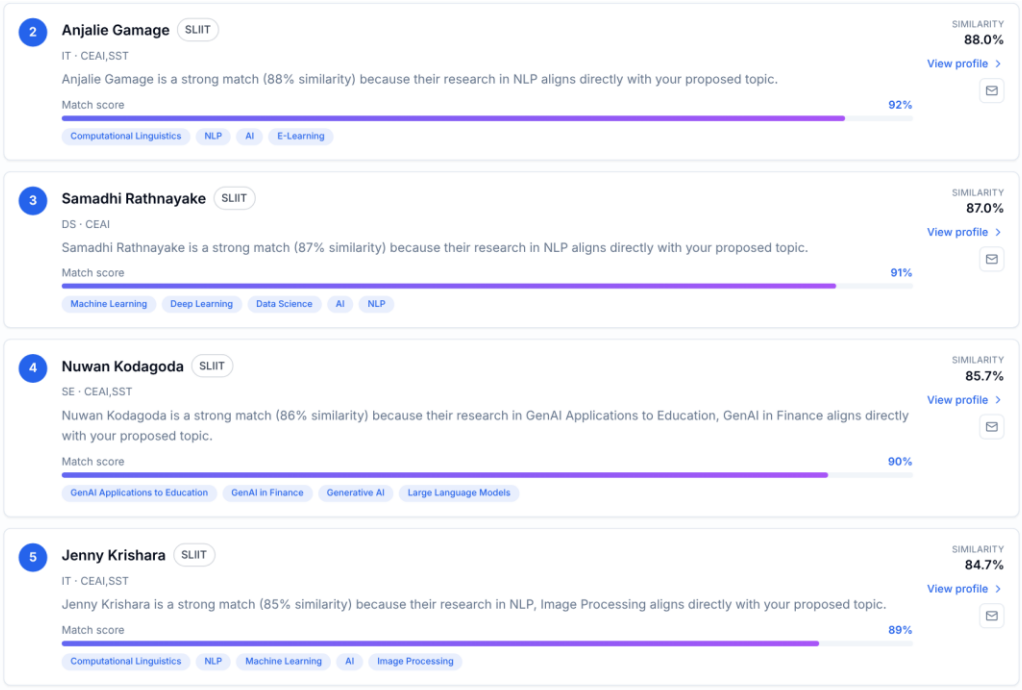


Figure 5.2: Remaining top-five supervisor matches for an NLP sentiment-analysis proposal

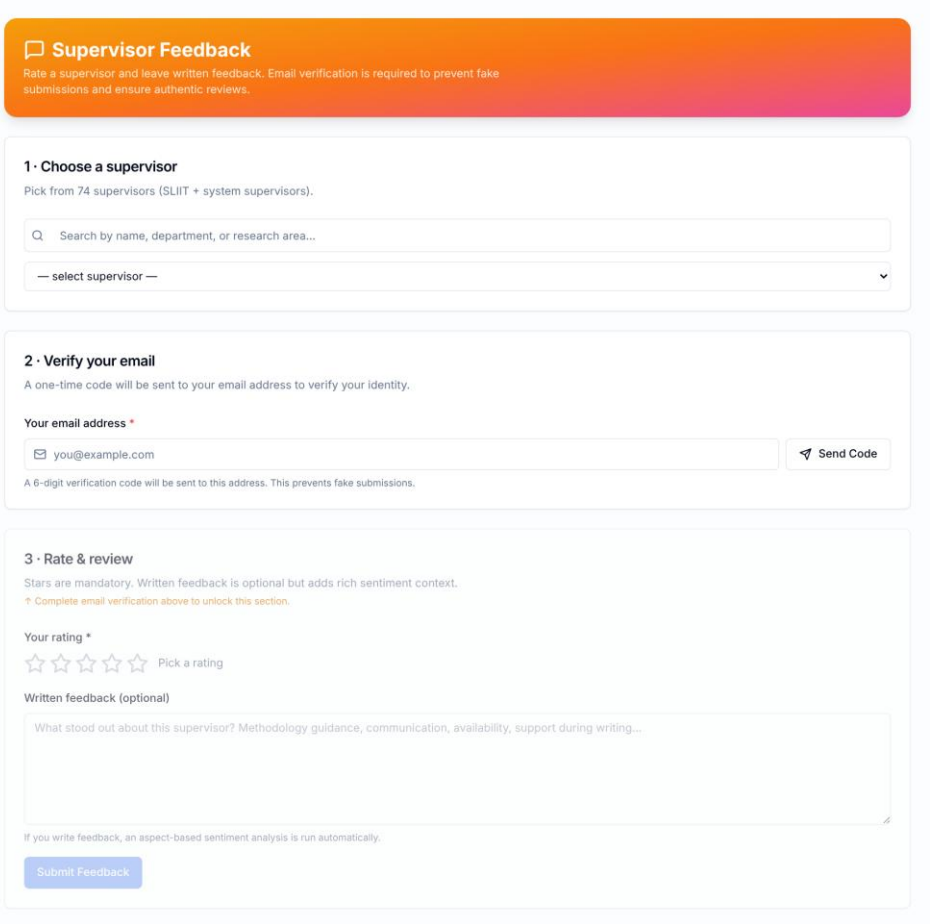


Figure 5.3: Supervisor Feedback page: supervisor selection, e-mail verification and rating steps

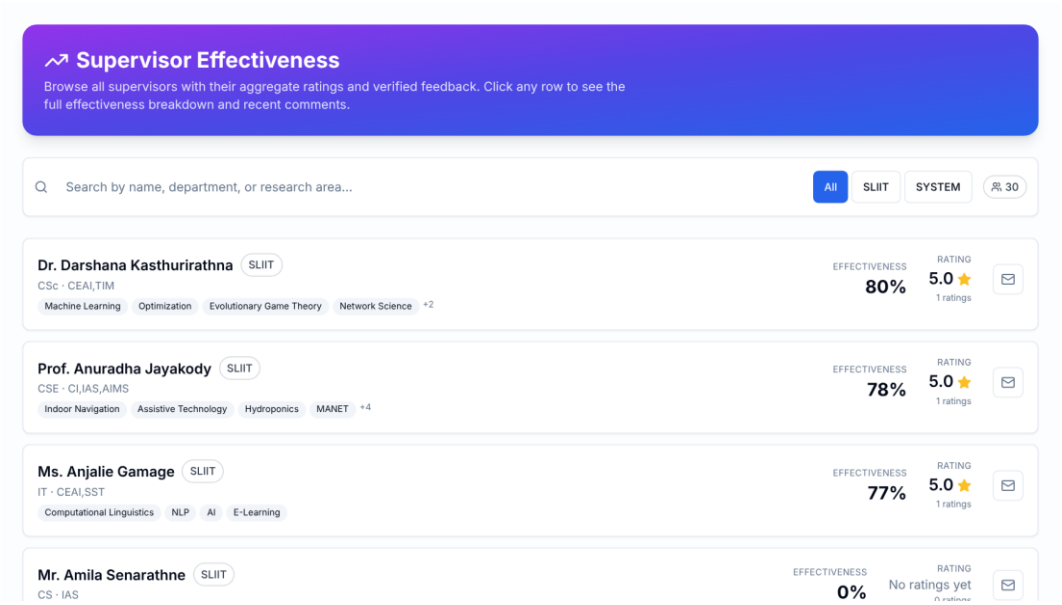


Figure 5.4: Supervisor Effectiveness page listing supervisors with effectiveness score and rating

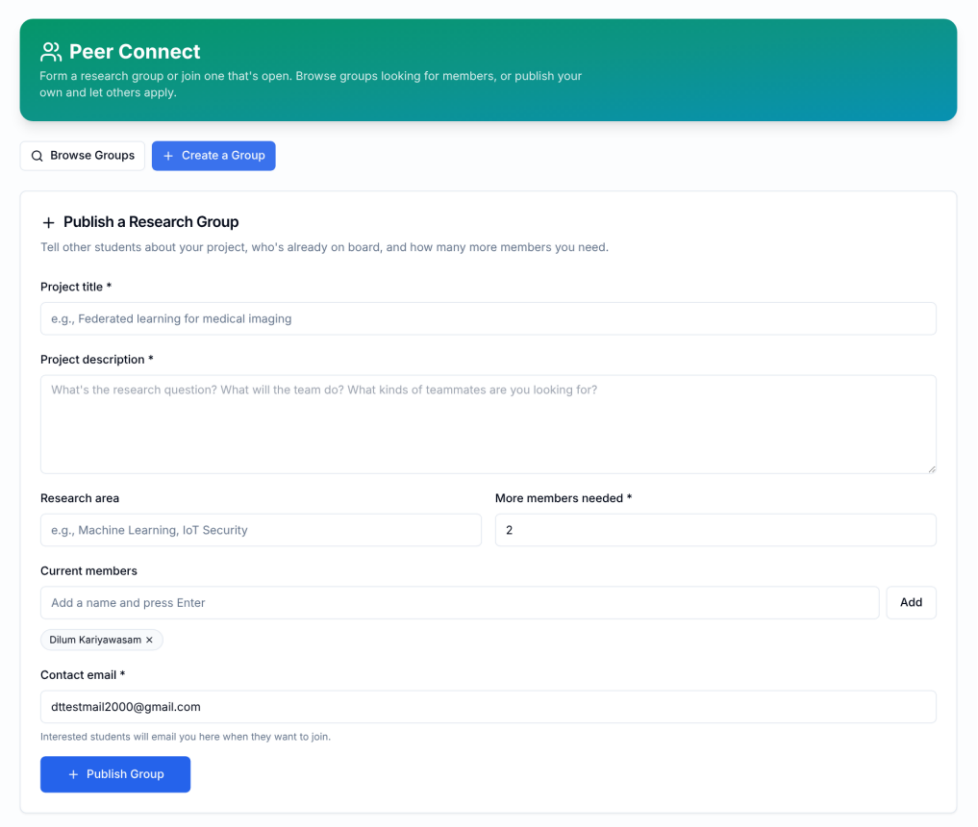


Figure 5.5: Peer Connect: form for publishing a research group with open slots

5.3 Integration with the Other Components

- **Module 1:** a topic refined through gap analysis or a generated proposal can be used directly as the matching query.
- **Module 3:** the supervisor matcher and Module 3 share the MiniLM embedding family; corpus duplicates detected by Module 3 also appeared in Semantic Scholar profile data.
- **Module 4:** proposals refined after supervisor matching can be scored for quality and success probability; completion rates recorded in `supervisor_matches` feed the effectiveness score.
- **Platform:** authentication, rate limiting and the dashboard shell are shared; Figure 5.6 shows the common request flow.

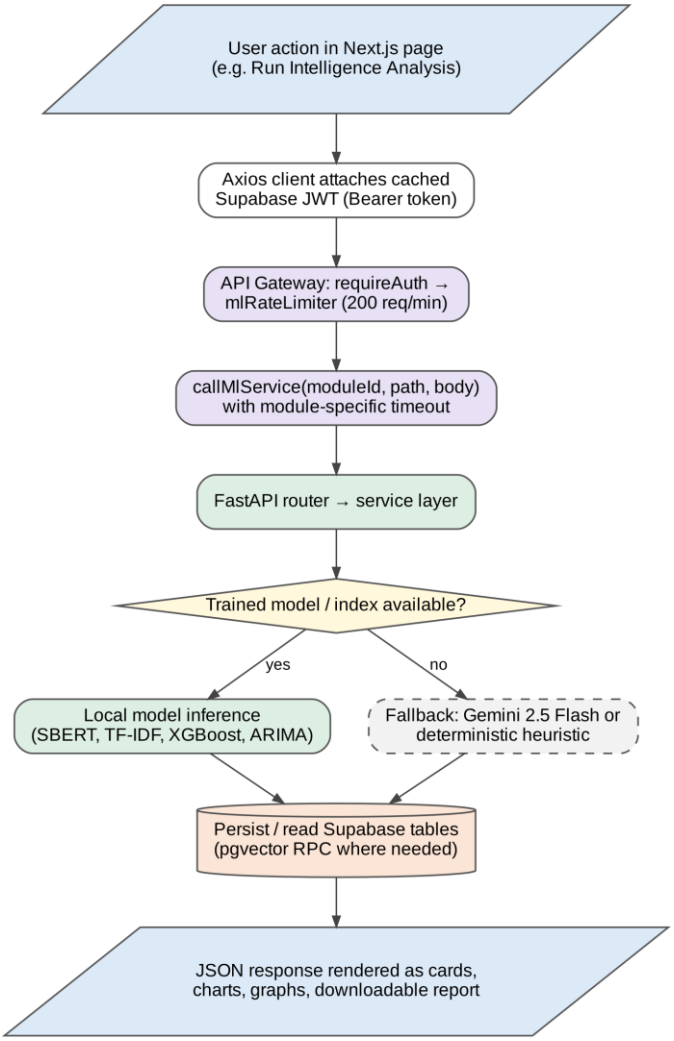


Figure 5.6: Request flow from a Next.js page through the gateway to a FastAPI service

5.4 Verification and Software Testing

Model-level evaluation is reported in Chapter 6. Functional validation executed 13 cases against the running service on 28 August 2026 (Table 5.3); the Playwright test `e2e/module2-supervisor.spec.ts` exercises the matching page end-to-end. Write endpoints (rating submission, group creation) were excluded from automated validation by team decision to avoid polluting production data and were verified manually.

Table 5.3: Functional validation of Module 2 (28 August 2026)

ID	Scenario	Result	Evidence or root cause
M2-01	Matching — CNN plant-disease proposal	PASS	All top-5 are Deep Learning / Computer Vision supervisors (similarity 0.74–0.77)

ID	Scenario	Result	Evidence or root cause
M2-02	Matching — network intrusion detection	PASS	All top-5 are cyber-security specialists (0.84–0.86)
M2-03	Matching — legacy interests + abstract (NLP)	PASS	All top-5 NLP / computational linguistics
M2-04	Matching — empty proposal	PARTIAL	Returns generic matches instead of a validation error
M2-05	Profile — supervisor 1	PASS	49 real papers 2003–2026; accurate AI focus summary
M2-06	Profile — supervisor 2	PARTIAL	50 papers correct; Gemini summary failed to parse
M2-07	Profile — invalid ID	PASS	Clear 404
M2-08/09	Groups — open / all	PASS	Three real groups listed; docstring path stale
M2-10	Sentiment — clearly positive	PASS	positive, overall 0.95, aspects differentiated
M2-11	Sentiment — clearly negative	PASS	negative, overall −0.95
M2-12	Sentiment — mixed feedback	FAIL	Gemini JSON parse failure → all aspects neutral, no error surfaced
M2-13	Rateable supervisor directory	PASS	SLIIT and system supervisors combined
M2-14	Effectiveness list	PASS	Formula hand-verified; SLIIT supervisors capped at 0.80

5.5 Deployment and Operation

The service is containerised (`services/module2-collaboration/Dockerfile`) with its `data/` directory copied into the image so that supervisor embeddings are available at start-up, and deployed on Railway with automatic redeployment from GitHub (Figure 5.7). Secrets (`SEMANTIC_SCHOLAR_API_KEY`, `GEMINI_API_KEY`, `SMTP_*`, Supabase keys) are injected as environment variables. Because fine-tuned weights are excluded from the repository for size, the deployed container loads the base `all-MiniLM-L6-v2` checkpoint unless the weights are supplied at build time; the minimum similarity was lowered to 0.2 on 13 May 2026 for compatibility with the base model.

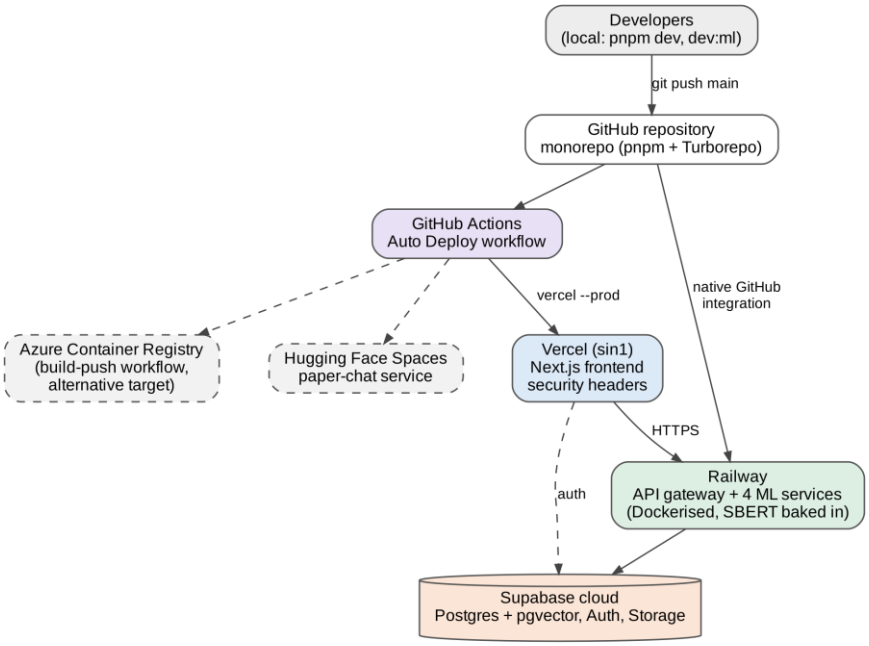


Figure 5.7: Continuous deployment pipeline of the platform

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 RQ1: Matching accuracy and ranking quality

During fine-tuning, validation cosine accuracy rose from 0.946 after the first epoch to 0.973 from the second epoch onwards, F1 reached 0.986, recall 1.000 and average precision 0.998 by epoch 12, with Matthews correlation 0.805 (Figure 6.1).

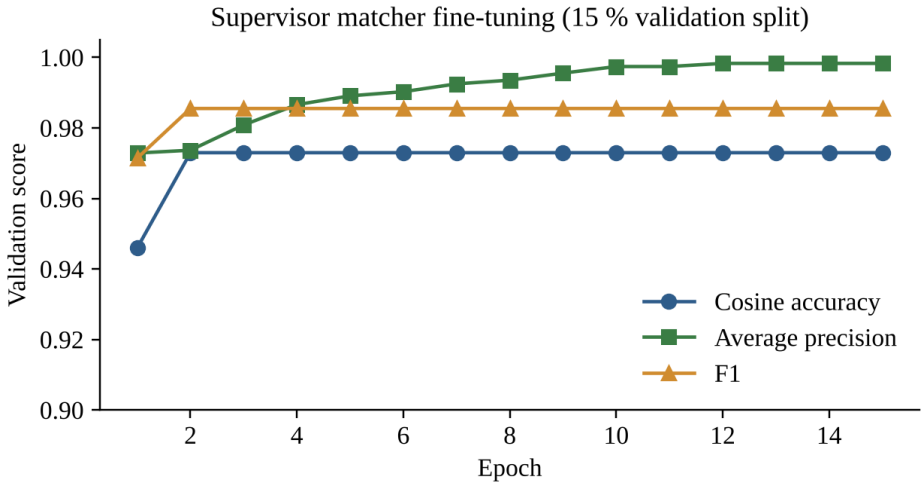


Figure 6.1: Validation metrics of the supervisor matcher during contrastive fine-tuning

Evaluated on all 147 labelled pairs at a cosine threshold of 0.60, the model achieved accuracy 0.946, precision 0.941, recall 1.000 and F1 0.970 (Figure 6.2), exceeding the 85% accuracy and 0.75 F1 targets. All eight errors were false positives — non-matching pairs scored above 0.60 — and no true match was missed.

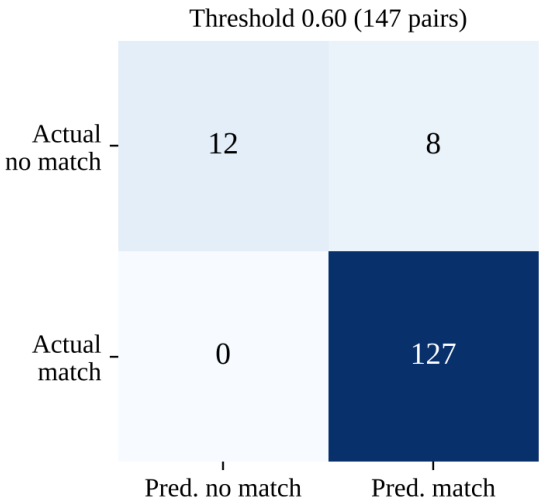


Figure 6.2: Confusion matrix of the supervisor matcher at threshold 0.60

Table 6.1 shows qualitative top-3 results for seven domain queries from the evaluation script. Rankings were topically correct for IoT agriculture, reinforcement learning, cyber security, e-learning, computer vision and big-data queries. For the NLP query, however, a medical-imaging supervisor ranked third (similarity 0.953), and all scores lie in a narrow band of 0.89–0.97, indicating that fine-tuning compressed the similarity scale.

Table 6.1: Qualitative top-3 matches from the evaluation script

Query (abridged)	Top-3 supervisors (declared interests)
NLP text classification with BERT	S. Rathnayake (ML, DL, Data Science) 0.959; S. Thelijjagoda (Computational Models of Human Language, NLP) 0.954; S. Chandrasiri (Medical Image Processing, AI) 0.953
IoT smart agriculture with sensor networks	P. Abeygunawardhana (Smart Agriculture) 0.958; V. Jayasinghearachchi (IoT, Edge, WSN) 0.948; N. Silva (IoT, Smart Homes) 0.934
Deep reinforcement learning for robotics	J. Perera (Deep RL) 0.941; J. Bamunusinghe (ML, AI) 0.897; P. Sumathipala (XAI, Evolutionary Computing) 0.886
Cyber security and intrusion detection	H. Fernando (Cyber Security, Networking) 0.950; D. Kasthurirathna (ML, Optimisation) 0.929; A. Senarathne (Information Security, Forensics) 0.928
E-learning and personalised education	K. Manathunga (Technology Enhanced Learning) 0.964; S. Kumari (ML, E-Learning) 0.963; A. Karunasena (ML, BI) 0.963
Computer vision and medical image analysis	L. Abeywardhana (DL, Computer Vision) 0.969; L. Weerasinghe (CV, Image Processing) 0.957; T. Shyamalee (ML, HCI, CV) 0.947
Big-data analytics and BI dashboards	T. Dassanayake (Big Data, BI) 0.971; S. Kasthuriarachchi (Big Data Analytics) 0.945; C. Kahandawaarachchi (Health Informatics, Big Data) 0.935

In the live validation (Section 5.4), three domain proposals — plant-disease detection with CNNs, network intrusion detection and NLP sentiment analysis — each returned five topically appropriate supervisors, a precision at five of 1.0 by evaluator judgement, with similarities of 0.74–0.88, a wider and more interpretable range.

6.1.2 RQ2: Zero-shot aspect sentiment

For unambiguous text the Gemini analyser was well calibrated: clearly positive feedback received an overall score of 0.95 and clearly negative feedback −0.95, with differentiated aspect probabilities (M2-10, M2-11). For mixed feedback — positive about methodology but critical of writing — the call failed to return parsable JSON and the system silently fell back to a uniform neutral result for all four aspects (M2-12). Two of roughly five Gemini-dependent calls failed in the validation session. RQ2 is therefore answered partially: zero-shot ABSA works for clear cases but is not yet reliable for the mixed feedback it is meant to handle, consistent with reports that LLMs lag on fine-grained sentiment tasks [22].

6.1.3 RQ3: Effectiveness scoring

The effectiveness formula was hand-verified against live data: the top supervisor (average 5.0 stars from one rating) scored 0.80. The verification also revealed a structural bias: SLIIT-directory supervisors have no match-tracking rows, so their completion term is always zero and, after re-normalisation, their maximum achievable score is about 0.80, whereas system-registered supervisors can reach 1.0.

6.2 Product and Technical Results

Table 6.2: Feature completion status of Module 2

Feature	Status	Notes
Supervisor matching + explanations	Complete	Fine-tuned model or base fallback
Supervisor profile (Semantic Scholar + Gemini)	Complete	Rate-limited, cached; summary best-effort
Peer Connect groups and join requests	Complete	mailto-based contact
Legacy SBERT peer matcher	Complete	Requires embedded profiles
OTP-verified ratings	Complete	SMTP or dev fallback
Aspect-based sentiment	Partial	Silent neutral fallback on parse failure

Feature	Status	Notes
Supervisor effectiveness	Complete	Structural cap for SLIIT-directory supervisors
Fine-tuned BERT sentiment model	Not trained	Wrapper only (app/models/sentiment.py)

Non-functional evidence: matching requires a single 384-dimensional encoding and 74 dot products, well within the 10-second target; profile calls respect the external rate limit through locking and caching; all routes are authenticated at the gateway; and invalid inputs return clear errors (M2-07).

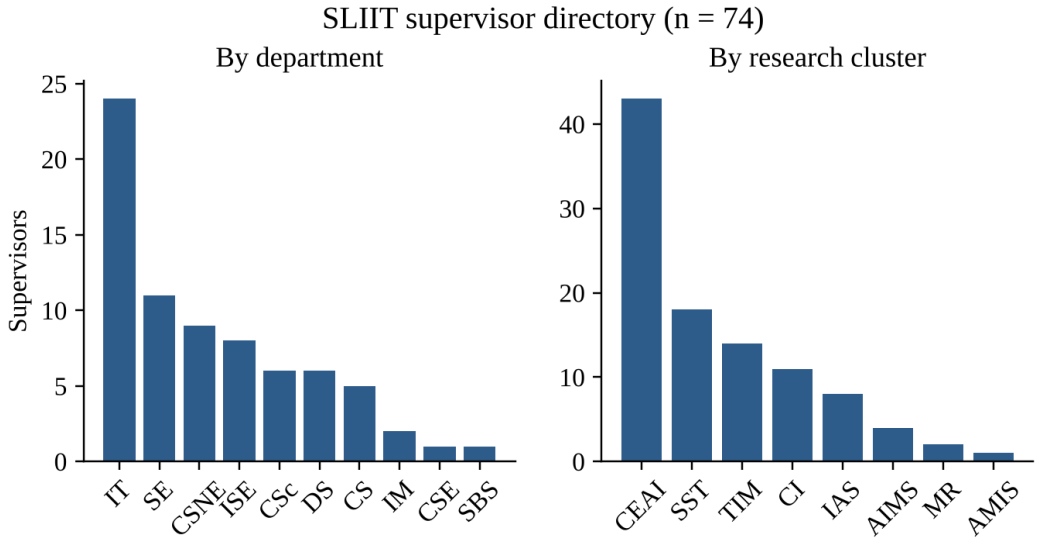


Figure 6.3: Distribution of the 74 supervisors in the matching directory by department and research cluster

6.3 User and Stakeholder Validation

The supervisor reviewed the module validation report and signed it off, noting that supervisor matching was the strongest feature and that the sentiment parse failure should be resolved before feedback is relied upon for real reviews. A task-based user-acceptance protocol for students is given in Appendix D.

6.4 Integration Results

All 17 gateway routes for Module 2 are reachable with authentication; the four collaboration pages share the dashboard shell and theme; peer groups, ratings and matches are persisted in Supabase with Row-Level Security.

6.5 Summary of Findings

Supervisor matching exceeded its accuracy and F1 targets on labelled pairs and produced topically correct rankings in live tests; profile enrichment adds real publication evidence; the group marketplace and verified ratings work end-to-end; but aspect sentiment fails silently on mixed feedback and the effectiveness score is structurally capped for directory supervisors. Overall, 10 of 13 validation cases passed, 2 were partial and 1 failed (Figure 6.4).

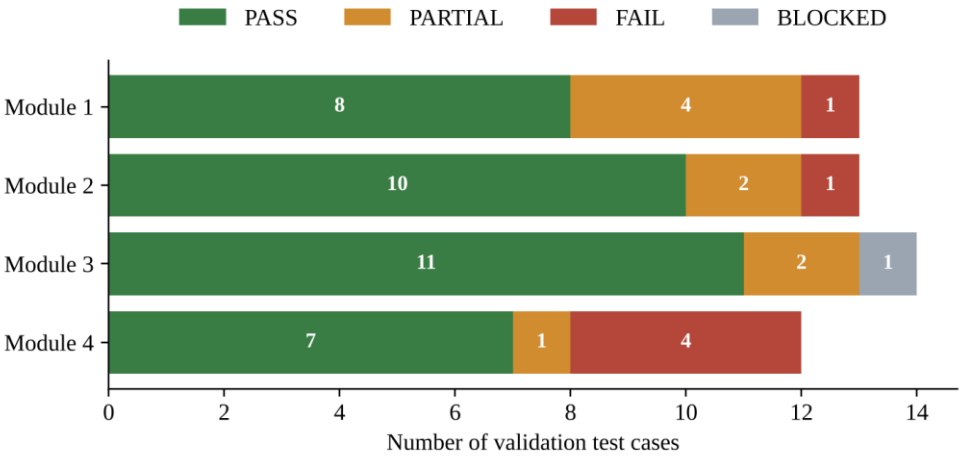


Figure 6.4: Validation outcomes per module (Module 2 in the second row)

7 DISCUSSION

7.1 Interpretation of Findings

The bi-encoder design delivered what matching needs most: correct supervisors at the top of the list at negligible query cost. The perfect recall and the false-positive-only error pattern mean that the model errs on the side of suggesting plausible supervisors rather than hiding suitable ones, which is acceptable for an advisory tool whose final decision rests with staff. However, the narrow similarity band after fine-tuning (0.89–0.97) shows that the contrastive objective with few negatives pulled most supervisors toward every query; absolute similarity values are therefore less informative than ranks, and the explanation thresholds (0.55, 0.70) were calibrated for the base model's wider range.

7.2 Comparison with Literature Baselines and Existing Solutions

Compared with collaborator recommenders that require co-authorship networks [6], [7], the module works for students with no publications because it relies only on free text. Compared with analytics-based supervisor selection [10], it adds semantic matching of proposals, capacity awareness and explanations. Compared with teaching-evaluation scores criticised as opaque [17], the effectiveness score exposes its components. No controlled comparison with manual allocation was performed.

7.3 Research Contribution

- A practical, explainable supervisor-matching design for undergraduates that blends semantic similarity with capacity and is evaluated on SLIIT data.
- Evidence that small-sample contrastive fine-tuning improves classification metrics but compresses the similarity scale, with implications for threshold-based explanations.
- An empirical observation of silent failure modes of zero-shot LLM aspect sentiment on mixed academic feedback.

7.4 Product Contribution

Students can move from a topic to a short, justified list of supervisors with publication evidence in seconds, form groups without a server-side mail relay, and give verified, aspect-level feedback. Administrators gain a transparent view of supervision quality.

7.5 Limitations and Threats to Validity

- **Evaluation leakage:** the 147-pair evaluation overlaps with the training data, and the set is imbalanced (127 positive, 20 negative); the 0.946 accuracy is optimistic.
- **Deployed model:** fine-tuned weights are not in the repository, so production may run the base checkpoint; live validation scores are consistent with the base model.
- **Directory currency:** interests are self-declared and may be outdated; Semantic Scholar name matching can merge or split authors.
- **Sentiment reliability:** Gemini parse failures produce silent neutral results; no labelled dataset exists to measure accuracy.
- **Effectiveness bias:** completion is unavailable for SLIIT-directory supervisors, capping their scores; one rating can dominate a score.
- **Edge cases:** an empty proposal returns generic matches; peer-router documentation lists a stale path.

7.6 Implications

For institutions, semantic matching can make allocation fairer to students without personal networks while keeping staff in control. For research, evaluation of people-recommenders should include ranking correctness and calibration of scores, not only binary accuracy. For responsible AI, generative components must surface failures rather than degrade silently.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

Primary users are students seeking supervisors and collaborators; **secondary users** are research coordinators and heads of department who allocate supervision and monitor its quality. The value lies in faster, better-justified matches, a transparent feedback loop and reduced administrative effort in allocation.

8.2 Market and Competitive Position

Researcher networks and staff directories support browsing but not matching; course evaluation systems collect ratings without aspect analysis (Table 2.2). The module is differentiated by proposal-level semantic matching, capacity awareness, publication evidence and aspect-level, verified feedback, and it becomes more valuable inside the integrated platform, where topics and proposals flow in from Module 1.

8.3 Business and Sustainability Model

Table 8.1: Indicative revenue model (from the approved proposal)

Phase	Strategy	Indicative pricing
Year 1	Free pilot at SLIIT; refine matching with real allocations	—
Year 2	Freemium: basic matching free; profile analytics and effectiveness dashboards premium	LKR 400 per user per month
Year 3	Institutional licence with directory onboarding and analytics	LKR 180,000 per institution per year

Costs are dominated by hosting (CPU inference) and Gemini calls for summaries and sentiment, which are cached (24 hours for focus summaries) to keep usage low.

8.4 Scalability, Intellectual Property and Compliance

- Matching cost grows linearly with the number of supervisors (one dot product each); thousands of supervisors remain trivial, and pgvector can index larger directories [35].

- The multi-factor policy, explanation templates, effectiveness formula and SLIIT supervisor dataset are the team's work; libraries are used under open-source licences.
- Ratings and e-mail addresses are personal data under the PDPA [26]; consent, retention limits and a right to contest scores are required before institutional roll-out.

8.5 Contribution to the Sustainable Development Goals

The platform contributes primarily to **SDG 4 (Quality Education)**, in particular target 4.3 (equal access to quality tertiary education) and target 4.4 (relevant skills), by giving every student — not only those with well-connected supervisors or paid tools — structured support for research. It also contributes to **SDG 9 (Industry, Innovation and Infrastructure)**, target 9.5 (enhance scientific research and upgrade technological capabilities), by turning an institutional repository into queryable research infrastructure [37]. Module 2 contributes in particular by making supervisor access depend on research fit rather than personal networks (supporting equitable access under SDG 4) and by fostering collaboration partnerships within the institution (SDG 17, partnerships).

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the Collaboration and Recommendation Engine of Researchly AI. A fine-tuned Sentence-BERT bi-encoder with a capacity-aware multi-factor score produced accurate, explainable supervisor recommendations; profile enrichment grounds them in real publications; a group marketplace and OTP-verified ratings support collaboration and feedback; and a transparent effectiveness score summarises supervision quality. The evaluation also identified limitations — optimistic evaluation, compressed similarity scores, unreliable zero-shot sentiment on mixed feedback and a structural score cap — that define the next steps.

9.2 Achievement of Objectives

Table 9.1: Achievement of objectives

Objective	Status	Evidence
SO1 Semantic supervisor matching (accuracy ≥ 85%, F1 ≥ 0.75)	Achieved	Section 6.1.1; Figures 6.1, 6.2
SO2 Peer collaboration feature	Achieved	Section 4.4.3; M2-08/09
SO3 Aspect-based feedback sentiment	Partly achieved	Section 6.1.2; M2-10 to M2-12
SO4 Supervisor-effectiveness score	Achieved (with known bias)	Section 6.1.3; M2-14
SO5 Multi-factor, explained ranking	Achieved	Section 4.4.1
SO6 Platform integration	Achieved	Section 6.4

9.3 Individual Research and Product Contributions

- Compiled the 74-supervisor dataset and 147 labelled pairs; fine-tuned and evaluated the matcher.
- Designed the multi-factor ranking and explanation logic.
- Implemented rate-limited, cached Semantic Scholar enrichment and Gemini focus summaries.
- Built the group marketplace, OTP-verified rating flow, aspect sentiment and effectiveness scoring, and their user interfaces.

9.4 Future Directions

- Evaluate on a held-out, balanced pair set and on real allocation outcomes; recalibrate explanation thresholds.
- Ship fine-tuned weights through a model registry and record the active model version in responses.
- Retry or surface Gemini parse failures; label a SLIIT feedback dataset and fine-tune a BERT ABSA model [15].
- Add completion tracking for directory supervisors and minimum-rating thresholds to the effectiveness score.
- Validate empty or very short proposals and correct the peer-router documentation.
- Run the student user-acceptance study in Appendix D.

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GLOSSARY

Term	Definition
Bi-encoder	A model that embeds each text separately so similarities can be computed between pre-computed vectors.
Contrastive loss	A training objective that pulls matching pairs together and pushes non-matching pairs apart in embedding space.
Multi-factor score	The ranking score combining semantic similarity and supervisor availability.
Aspect	A dimension of supervision (methodology, writing, originality, data analysis) whose sentiment is analysed separately.
Re-normalisation	Dividing by the sum of weights of available signals so missing signals do not count as zero.
Hard negative	A non-matching example that is superficially similar, used to sharpen a model's decision boundary.

APPENDICES

Appendix A: Extended Results and Test Evidence

Table A.1: Supervisor matcher fine-tuning configuration

Parameter	Value
Base model	sentence-transformers/all-MiniLM-L6-v2 (384-d)
Loss	Contrastive loss with synonym substitution and hard negatives
Epochs / batch size	15 / 16
Learning rate / warm-up	2×10^{-5} / 10 steps
Validation split / seed	15% / 42
Final validation metrics	accuracy 0.973, F1 0.986, precision 0.971, recall 1.000, AP 0.998, MCC 0.805

Recorded response excerpt for validation case M2-02 (network intrusion detection proposal):


```
{ "supervisor_id": 71, "name": "Uditha Dharmakeerthi", "department":
"CSNE",
  "research_interests": ["Computer Networks", "Cyber Security"],
  "similarity_score": 0.8597, "multi_factor_score": 0.9018,
  "explanation": "... is a strong match (86% similarity) because their
research
                  in Computer Networks, Cyber Security aligns directly
..." }
```

Appendix B: Academic Integrity and AI-Use Disclosure

Table B.1: AI-use disclosure

Tool	Purpose	Verification performed	Sections affected
Claude (Anthropic) — AI coding assistant	Debugging suggestions; execution of the module validation harness and drafting of the module report (28 Aug 2026)	Evidence inspected; supervisor sign-off obtained	Chapter 5.4, Chapter 6
Claude (Anthropic)	Drafting and formatting this thesis from repository evidence; diagrams and charts from repository data	Numbers cross-checked with repository JSON/CSV; references checked	All chapters
Google Gemini 2.5 Flash	Runtime component (sentiment, focus summaries)	Outputs treated as untrusted; fallbacks implemented	Chapter 4

Appendix C: Additional Interface Screenshots

Screenshots were captured from the production build of the Researchly AI front end rendering responses recorded from the live Module 2 service during the 28 August 2026 validation run.

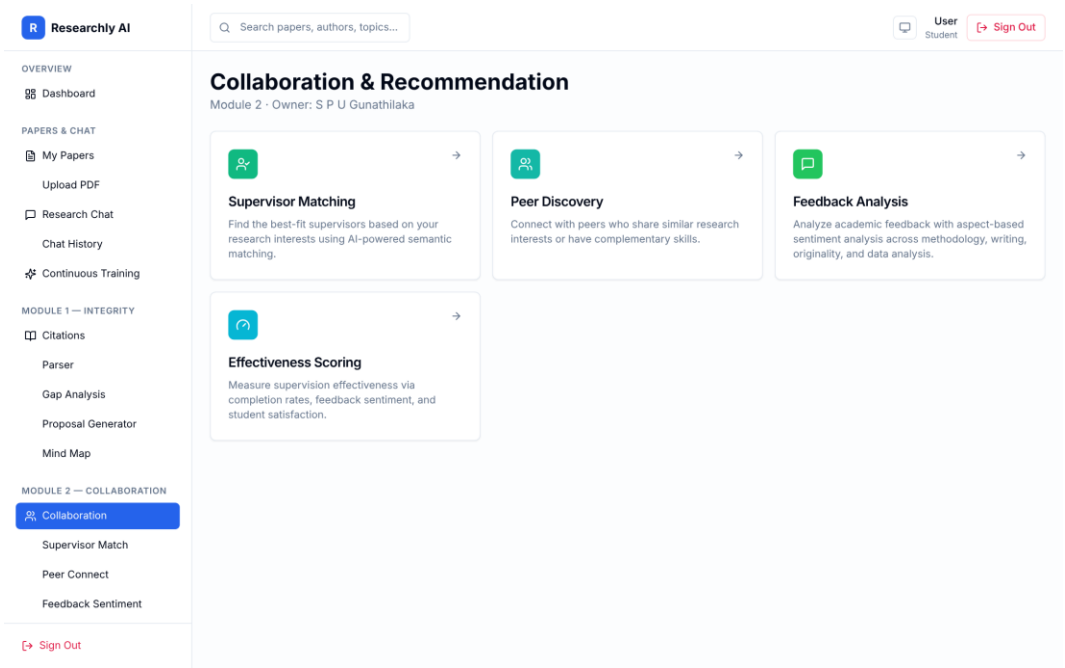


Figure C.1: Collaboration and Recommendation overview page linking the four Module 2 features

Appendix D: User Acceptance Testing Protocol

Participants complete the following tasks unaided; completion, time and errors are recorded and the System Usability Scale [38] is administered afterwards.

Table D.1: UAT tasks and recording sheet for Module 2

Task	Expected outcome	Completed (Y/N)	Time (s)	Errors / comments
T1 Find three suitable supervisors for your own proposal	Ranked list with reasons			
T2 Open a supervisor profile and identify their recent focus	Profile, charts, summary			
T3 Publish a research group with two open slots	Group visible in Browse			
T4 Request to join another group	Pre-filled e-mail opens			
T5 Verify your e-mail and rate a supervisor with written feedback	Rating with aspects saved			
T6 Explain why a supervisor has their effectiveness score	Breakdown read correctly			